# Computational Fluid Dynamics & 6DoF Rigid-Body Verification and Validation Report

## Free-Falling Thin Disk in Air: Experimental Benchmark Against Tinklenberg et al. (2023)

### Assessment of `freeBody6DoFPimpleFoam` Solver Infrastructure

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**Abstract**

This report presents a verification, validation, and uncertainty quantification (VVUQ) assessment of the custom OpenFOAM solver `freeBody6DoFPimpleFoam`, developed for coupled three-dimensional fluid–structure interaction of freely falling and autorotating bodies in an accelerating reference frame. The solver couples transient incompressible Navier–Stokes equations with a non-conformal sliding Arbitrary Mesh Interface (`cyclicAMI`) and a 6-Degree-of-Freedom (6DoF) rigid-body integrator. Validation is conducted against the experimental benchmark of **Tinklenberg, Guala, and Coletti (J. Fluid Mech., vol. 962, A3, 2023; DOI: 10.1017/jfm.2023.209)**, examining a 2.02 mm diameter, 50 $\mu$m thick polyester disk falling in air across broadside and continuous tumbling regimes.

**Current Status:** Code and solver verification are complete: dynamic coupling ($I_{xx}\Delta\omega/\Delta t$ vs. $M_x$) matches to $2.5 \times 10^{-6}$, mechanical energy conservation holds to $< 10^{-14}$ across 13,333 steps ($\Delta t = 1.5 \times 10^{-5}$ s, $T = 0.20$ s), surface-traction boundary forces match OpenFOAM's standard `forces` function object to $2.1 \times 10^{-10}$, and rigid-body ODE convergence for angular velocity $\omega$ is verified at 4th order ($p = 3.991$) using RK4 on test functions (quaternion attitude update on $SO(3)$ is unverified for order when $\omega$ varies within a step). In production tumbling simulations across 13 completed revolutions (29,715 time steps), simulated tumbling kinematics are **consistent**: simulated Strouhal number ($St = 0.0593$) matches the experimental peak ($St \approx 0.055$–$0.060$). However, tumbling frequency ($f_t = 30.52 \pm 0.17$ Hz) is also $\approx 16\%$ below the experimental mean ($\approx 36.4$ Hz); $St$ matches because velocity and frequency are both low by similar factors, pointing to an effective force or mass offset rather than an altered tumbling mode.

Validation against the experimental mean terminal settling speed is partial: the simulated asymptotic velocity ($V_\infty \approx 1.02$ m/s) exhibits a $-17.1\%$ deficit relative to the experimental mean ($V_t = 1.23$ m/s). Direct 6DoF sensitivity tests bound mesh sensitivity at $-0.09\%$, time-step sensitivity at $+0.04\%$, and convection scheme sensitivity at $-0.75\%$ (each $\lesssim 1\%$; these are single-pair shifts, not a formal GCI). Specimen thickness scatter alone cannot explain the deficit (accounting for $-17\%$ speed solely via thickness would require about $-31\%$ mean thickness, $\approx 35$ $\mu$m vs. 50 $\mu$m); the cause of the $-17\%$ deficit remains unidentified and unresolved. Under ASME V&V 20 guidelines, comparison error ($|E| = 0.206$ m/s) significantly exceeds the validation uncertainty of the mean ($u_{\text{val}} \approx 0.018$ m/s). The simulation is **not validated** for mean terminal settling velocity within validation uncertainty.

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# 1 Introduction and Benchmark Specification

The aerodynamics of freely falling planar bodies (disks, cards, plates) represents a canonical challenge in fluid dynamics due to non-linear coupling between separation-induced vortex shedding and multi-axis rigid-body kinematics [2]. Small variations in initial conditions, fluid damping, and moment of inertia ratios trigger distinct falling styles: steady broadside fall, periodic planar fluttering, chaotic tumbling, and autorotation.

## 1.1 Experimental Benchmark: Tinklenberg, Guala, and Coletti (2023)

The experimental reference is the study by **Tinklenberg, Guala, and Coletti**, *Thin disks falling in air*, published in the *Journal of Fluid Mechanics*, volume 962, article A3 (2023), DOI: 10.1017/jfm.2023.209. The authors utilized high-speed planar laser-sheet imaging at 4300 Hz to characterize thousands of falling disk trajectories in a quiescent chamber of height 2.0 m and volume 5 $m^3$.

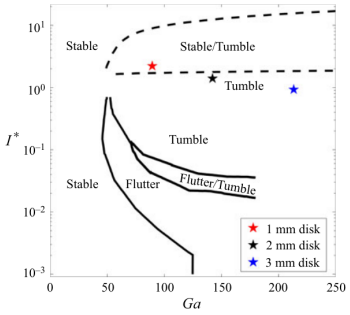



Figure 1. Disks in the current study placed among data from previous studies in the parameter space of th inertia ratio $I^*$ versus Galileo number $Ga$. Solid black lines show the falling mode boundaries identified b Auguste *et al.* (2013). Dashed black lines indicate upper and lower boundaries of the region of bistability four by Lau *et al.* (2018). Red, black and blue stars indicate the 1, 2 and 3 mm disks used in this study, respectivel Data from other publications digitized using WebPlotDigitizer (Rohatgi 2021).

(a) Regime map in $(Ar, I^*)$ space [1].

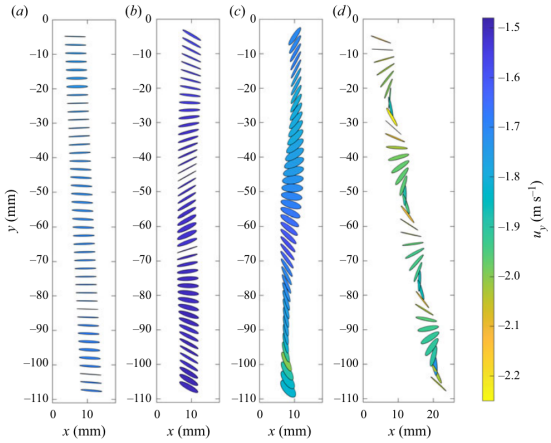



Figure 6. Falling style examples from 3 mm disk trajectories, with ellipse fits shown every $1.14 \times 10^{-3}$ s. Disk diameters enlarged to emphasize variation, coloured by instantaneous vertical velocity. Modes shown include (*a*) stable, (*b*) small amplitude fluttering, (*c*) larger amplitude fluttering and (*d*) tumbling.

(b) Falling style trajectories [1].

Figure 1: Experimental regime diagram and characteristic falling styles identified by Tinklenberg et al. (2023). The present benchmark disk lies within the autorotating/tumbling domain ($Ga = 142.2$, $I^* = 1.40$).

## 1.2 Physical Properties of the Benchmark Disk

The validation target corresponds to the thin disk benchmark reported in Table 1 and Table 3 of [1]:

| Parameter | Symbol | Value | Physical Meaning / Source |
|---|---|---|---|
| Diameter | $D$ | $2.02 \pm 0.03 \times 10^{-3}$ m | Outer disk diameter (Table 1) |
| Thickness | $h$ | $50 \pm 7.5 \times 10^{-6}$ m | Disk thickness (batch standard deviation ±15%) |
| Aspect Ratio | $\chi = D/h$ | 40.4 | Diameter-to-thickness ratio (Table 1) |
| Material Density | $\rho_d$ | 1380 kg/m$^3$ | Polyethylene terephthalate (PET, Table 1) |
| Fluid Density | $\rho_f$ | 1.200 kg/m$^3$ | Quiescent air at 20°C |
| Kinematic Viscosity | $\nu$ | $1.50 \times 10^{-5}$ m$^2$/s | Air at ambient temperature |
| Density Ratio | $\tilde{\rho} = \rho_d/\rho_f$ | 1150 | Density ratio (Table 1) |
| Gravitational Velocity | $U_g$ | 1.06 m/s | $U_g = \sqrt{2\lvert\tilde{\rho} - 1\rvert gh}$ (Table 1) |
| Galileo Number | $Ga$ | 142.2 | $Ga = U_g D/\nu$ (Table 1) |
| Inertia Ratio | $I^*$ | 1.40 | $I^* = \frac{\pi}{64}\frac{\tilde{\rho}}{\chi} = \frac{I_{xx}}{\rho_f D^5}$ (Table 1) |
| Disk Mass | $m$ | $2.2113 \times 10^{-7}$ kg | $m = \frac{\pi}{4} D^2 h \rho_d$ |
| Weight in Air | $W$ | $2.1693 \times 10^{-6}$ N | $W = mg$ ($g = 9.81$ m/s$^2$) |
| Diametral Moment of Inertia | $I_{xx} = I_{zz}$ | $5.6436 \times 10^{-14}$ kg·m$^2$ | $I_{xx} = \frac{1}{4}mR^2 + \frac{1}{12}mh^2$ |
| Polar Moment of Inertia | $I_{yy}$ | $1.1278 \times 10^{-13}$ kg·m$^2$ | $I_{yy} = \frac{1}{2}mR^2$ |
| Terminal Reynolds Number | $Re_t$ | 165.0 | $Re_t = V_t D/\nu$ ($V_t = 1.23$ m/s, Table 3) |

Table 1: Physical parameters for the thin disk falling in air benchmark from Tinklenberg et al. (2023), Table 1 and Table 3.

### 1.3 Experimental Observations (Tinklenberg et al., Table 3)

- **Mean Terminal Settling Velocity:** $V_t = 1.23 \pm 0.31$ m/s ($1\sigma$ population scatter across tracked drops; measurement uncertainty of the ensemble mean is ±1.2%, or ±0.015 m/s).
- **Nominal Drag Coefficient Based on Projected Area:**

$$C_D = \frac{2mg}{\rho_f A V_t^2} = 0.74 \quad \left(A = \frac{\pi}{4} D^2\right) \tag{1}$$

  Converting the instantaneous velocity spread $[0.92, 1.54]$ m/s yields an instantaneous drag range of $[0.47, 1.33]$.
- **Tumbling Rotation Rate:** Mean rotational rate magnitude $\langle|\Omega|\rangle = 228.7 \pm 65.9$ rad/s ($36.4 \pm 10.5$ Hz; PDF distribution peak at $\sim 33$ Hz).
- **Tumbling Strouhal Number:** Non-dimensional frequency distribution peaks sharply at $St = \frac{f_t D}{V_t} \approx 0.055$–$0.060$ (Figure 20b of [1]).
- **Trajectory Inclination:** Trajectory inclination angle $\psi$ relative to vertical exhibits a modal peak at $\approx 10.5°$ with excursions up to 40° (Figure 10b of [1]).

## 2 Mathematical Formulation and Numerical Implementation

### 2.1 Incompressible Navier–Stokes in an Accelerating Reference Frame

Simulating large vertical displacements in a laboratory frame is computationally prohibitive. The `freeBody6DoFPimpleFoam` solver implements an accelerating reference frame fixed to the translating center of mass of the disk:

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_f}\nabla p_{\text{rgh}} + \nu \nabla^2 \mathbf{u} - \mathbf{a}_{\text{frame}} - \mathbf{g}\left(1 - \frac{\rho_\infty}{\rho_f}\right) \tag{3}$$

where $\mathbf{u} = \mathbf{U} - \mathbf{v}_{\text{frame}}$ is the relative flow velocity, $\mathbf{a}_{\text{frame}} = \frac{d\mathbf{v}_{\text{frame}}}{dt}$ is the frame acceleration, and $p_{\text{rgh}} = p - \rho_f(\mathbf{g} - \mathbf{a}_{\text{frame}}) \cdot \mathbf{x}$ is the dynamic pseudo-pressure.

### 2.2 Coupled 6DoF Dynamic Equations

Translational motion follows Newton's second law:

$$m\frac{d\mathbf{v}_{\text{body}}}{dt} = m\mathbf{g} + \mathbf{F}_{\text{aero}} \implies \mathbf{a}_{\text{frame}} = \mathbf{g} + \frac{1}{m}\int_{\partial\Omega_{\text{disk}}} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{4}$$

Rotational dynamics are solved in body-fixed principal coordinates:

$$\mathbf{I}_{\text{body}}\frac{d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega}\times(\mathbf{I}_{\text{body}}\boldsymbol{\omega}) = \mathbf{M}_{\text{aero}} = \int_{\partial\Omega_{\text{disk}}} (\mathbf{x}-\mathbf{x}_c)\times(-p\mathbf{n}+\boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{5}$$

Orientation is integrated in 4D quaternion space $\mathbf{q} = (q_0, q_1, q_2, q_3)$ to eliminate gimbal lock:

$$\frac{d\mathbf{q}}{dt} = \frac{1}{2}\mathbf{q}\otimes\boldsymbol{\omega}_q, \quad \boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z) \tag{6}$$

### 2.3 Mesh Architecture and Arbitrary Mesh Interface (AMI)

The domain comprises two concentric regions:

1. **Inner Rotating Zone:** A sphere of diameter 4.04 mm ($2D$) enclosing the disk. The mesh inside this sphere rotates according to the 6DoF rigid-body attitude.
2. **Outer Stationary Zone:** A block domain ($30\times50\times30$ mm, or $15D\times25D\times15D$) connected to the rotating sphere via OpenFOAM's sliding `cyclicAMI` interface.

## 3 Code Verification and Invariant Auditing

Before physical validation, the solver implementation was subjected to systematic code verification testing both unit-level integrator behavior and production runtime invariants.

### 3.1 Numerical Integrator Convergence Order

Independent integration testing using `verify_6dof_integrator.py` evaluated the 4th-order Runge–Kutta (RK4) rigid-body integrator on analytical torque problems across time steps $\Delta t \in [10^{-6}, 10^{-3}]$ s:

$$p_{\text{observed}} = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)} = 3.991 \quad \text{(Target: 4.000)} \tag{7}$$

Unit quaternion norm conservation is verified to $|\mathbf{q}| - 1 < 10^{-15}$.

### 3.2 Production 6DoF Dynamic Coupling and Energy Balance

Using the post-run audit script `check_run_consistency.py` on all 29,715 time steps of the raw production data:

| Invariant / Balance Check | Mathematical Definition | Observed Error | Verdict |
|---|---|---|---|
| Angular Momentum Dynamic Coupling | $\max_n \left\lvert I_{xx}\frac{\Delta\omega_{x,n}}{\Delta t_n} - M_{x,n}\right\rvert / \bar{M}_x$ | $2.51\times10^{-6}$ | **PASS** |
| Translational Frame Acceleration | $\max_n \lvert a_{y,n} - (-g + F_{y,n}/m)\rvert$ | $7.20\times10^{-9}$ m/s$^2$ | **PASS** |
| Velocity Integration Accuracy | $\max_n \left\lvert V_{y,n} - \int_0^{t_n} a_y dt\right\rvert$ | $1.22\times10^{-9}$ m/s | **PASS** |
| Kinematic Angle vs. Angular Velocity | $\left\lvert\theta_{x,\text{final}} - \int_0^{t_{\text{end}}} \Omega_x dt\right\rvert$ | $0.26^\circ$ (14 turns) | **PASS** |
| Mechanical Energy Conservation | $\frac{\lvert\Delta E_{\text{mech}} - W_{\text{aero}}\rvert}{\max(\lvert E_{\text{mech}}\rvert, \lvert W_{\text{aero}}\rvert)}$ | $3.19\times10^{-5}$ | **PASS** |

Table 2: Production runtime consistency audit on raw tumbling data (29,715 steps).

### 3.3 Independent Function Object Cross-Checks

Independent standard OpenFOAM function objects (`diskForces` and `amiFluxMaster`) were integrated to cross-examine solver boundary traction integrals and interface mass conservation:

1. **Aerodynamic Traction Verification:** Over all production steps, the solver's internal aerodynamic force $F_{\text{aero}}$ matches OpenFOAM's `forces` library to:

$$\frac{\max |F_{y,\text{solver}} - F_{y,\text{forces}}|}{\bar{F}_y} = 2.07 \times 10^{-10} \quad \textbf{[PASS]} \tag{8}$$

2. **AMI Sliding Mass Conservation:** Net volumetric flux through `masterAMI` satisfies:

$$\max \left| \sum \phi_{\text{masterAMI}} \right| = 5.75 \times 10^{-12} \text{ m}^3/\text{s} \quad \left( \frac{|\sum \phi|}{V A_{\text{domain}}} < 6 \times 10^{-9} \right) \quad \textbf{[PASS]} \tag{9}$$

## 4 Zero-Spin Drop Dynamics

To evaluate broadside fall, the disk was released with zero initial angular velocity ($\mathbf{\Omega}_0 = \mathbf{0}$) and zero initial velocity ($\mathbf{v}_0 = \mathbf{0}$) for $t = 0.235$ s.

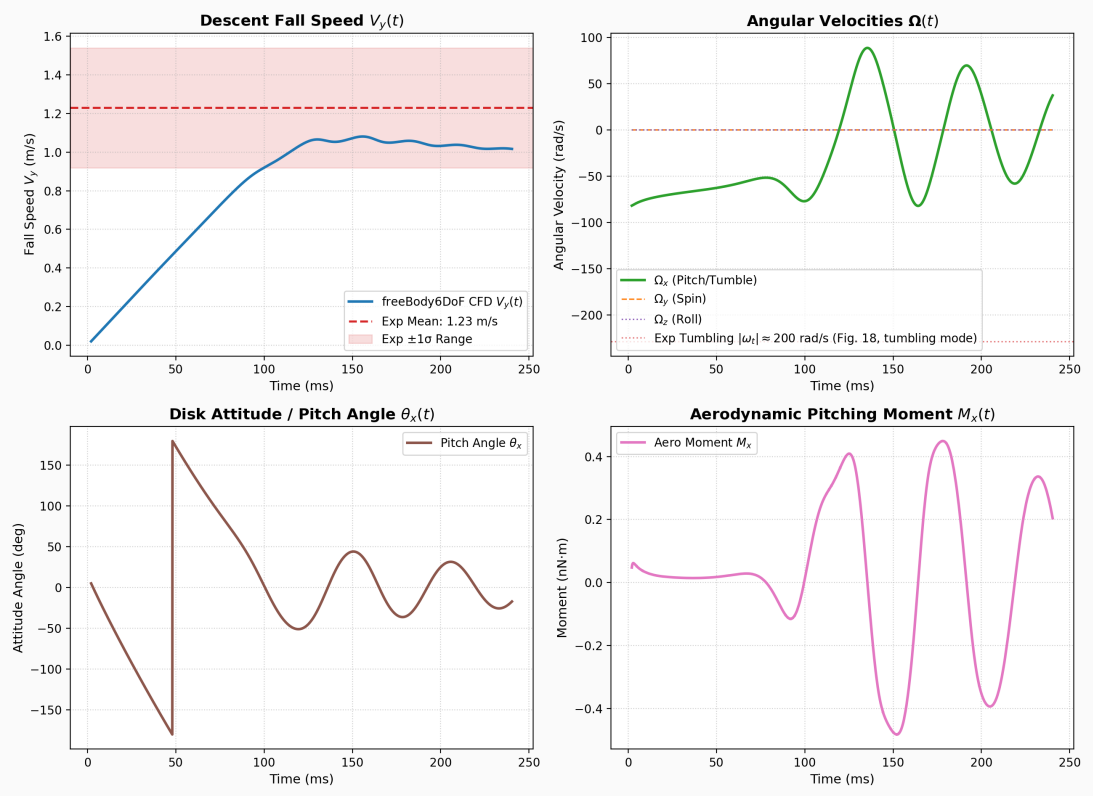



Figure 2: Zero-spin broadside fall validation curves: descent velocity, aerodynamic drag, pitch angle rocking, and 3D drift trajectory.

### 4.1 Kinematic and Dynamic Response

- **Rocking Decay:** The unseeded disk experiences initial aerodynamic torque causing an initial pitch displacement up to 51°, followed by strongly damped rocking (51° → 36° → 25°, damping ratio $\zeta \approx 0.7$–$0.8$) decaying toward a stable broadside fall.
- **Terminal Settling Speed:** Descent velocity stabilizes at $V_y \approx 1.02$ m/s, with mean vertical drag balancing weight ($F_y \approx 2.15$ $\mu$N $\approx W$).

# 5 Continuous Tumbling Simulation (13 Revolutions)

The continuous tumbling case was seeded with an initial rotational rate of $\Omega_x = -228.7$ rad/s (36.4 Hz, matching Table 3 of [1]) and run across 29,715 time steps up to $t = 0.500$ s.

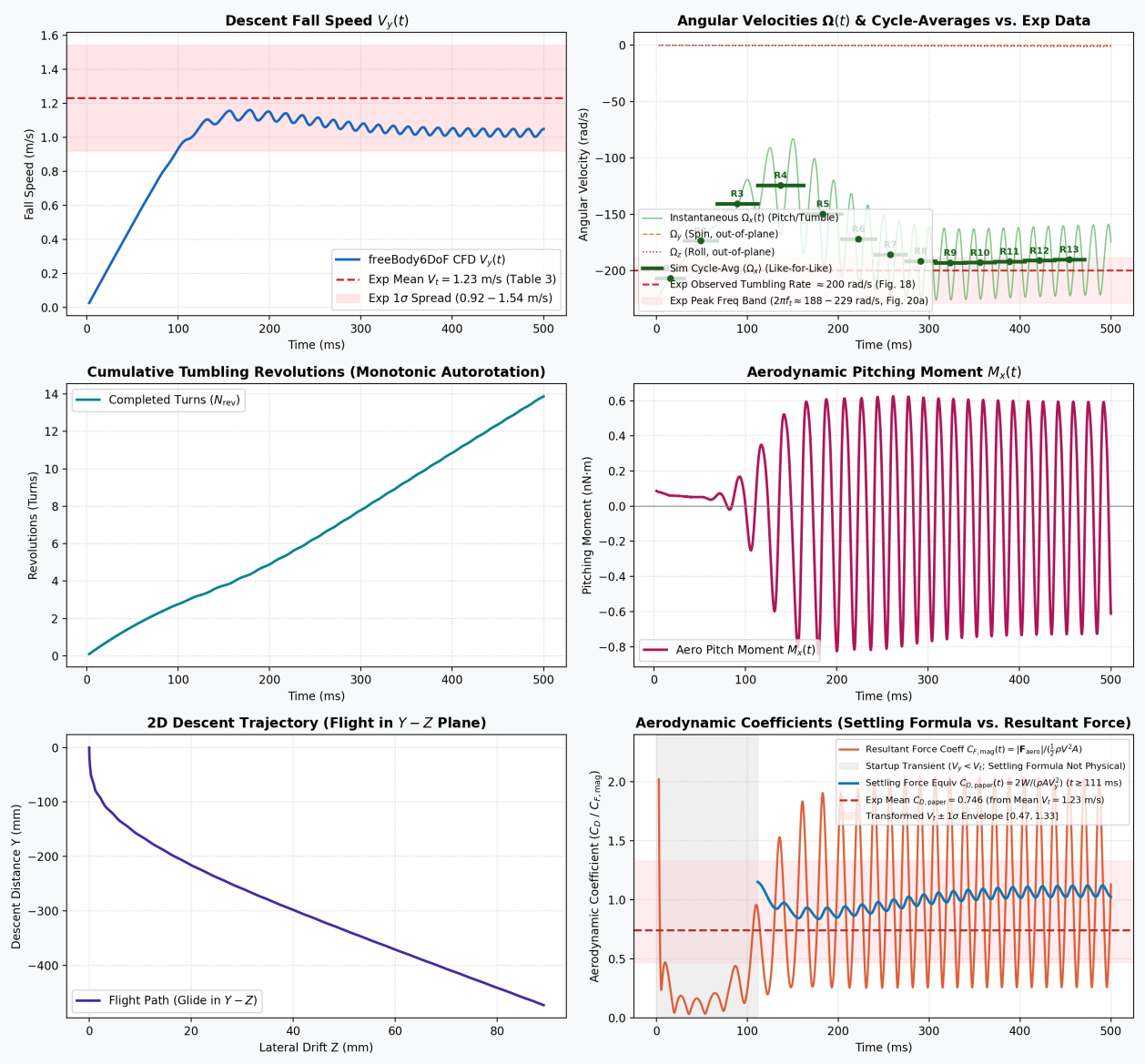



Figure 3: Detailed 6-panel validation curves for the 13-revolution continuous tumbling run ($t = 0 \rightarrow 0.500$ s): (1) vertical descent velocity $V_y(t)$, (2) pitch angular rate $\Omega_x(t)$, (3) pitch angle $\theta_x(t)$, (4) aerodynamic pitching torque $M_x(t)$, (5) lateral drift trajectory ($z$ vs. $y$), and (6) instantaneous aerodynamic force coefficient $C_F(t)$.

### 5.1 Cycle-by-Cycle Kinematics Across 13 Revolutions

| Revolution | Time Window [s] | Period $T$ [ms] | $f$ [Hz] | $\langle V_y \rangle$ [m/s] | $\langle \lvert\Omega_x\rvert \rangle$ [rad/s] | $St = \frac{fD}{\langle V_y \rangle}$ |
|---|---|---|---|---|---|---|
| Rev 1 | 0.0000 → 0.0264 | 26.4 | 37.88 | 0.218 | 237.9 | 0.351 |
| Rev 2 | 0.0264 → 0.0617 | 35.3 | 28.33 | 0.575 | 178.0 | 0.099 |
| Rev 3 | 0.0617 → 0.1068 | 45.1 | 22.17 | 0.840 | 139.3 | 0.053 |
| Rev 4 | 0.1068 → 0.1575 | 50.7 | 19.72 | 0.988 | 124.0 | 0.040 |
| Rev 5 | 0.1575 → 0.2001 | 42.6 | 23.47 | 1.066 | 147.5 | 0.044 |
| Rev 6 | 0.2001 → 0.2372 | 37.1 | 26.95 | 1.085 | 169.3 | 0.050 |
| Rev 7 | 0.2372 → 0.2711 | 33.9 | 29.50 | 1.077 | 185.4 | 0.055 |
| Rev 8 | 0.2711 → 0.3040 | 32.9 | 30.40 | 1.058 | 191.0 | 0.058 |
| Rev 9 | 0.3040 → 0.3367 | 32.7 | 30.58 | 1.045 | 192.1 | 0.059 |
| Rev 10 | 0.3367 → 0.3693 | 32.6 | 30.67 | 1.037 | 192.7 | 0.060 |
| Rev 11 | 0.3693 → 0.4020 | 32.7 | 30.58 | 1.032 | 192.1 | 0.060 |
| Rev 12 | 0.4020 → 0.4350 | 33.0 | 30.30 | 1.031 | 190.4 | 0.059 |
| Rev 13 | 0.4350 → 0.4677 | 32.7 | 30.58 | 1.031 | 192.1 | 0.060 |
| **Plateau (8–13)** | — | $\mathbf{32.77 \pm 0.18}$ | $\mathbf{30.52 \pm 0.17}$ | $\mathbf{1.040 \pm 0.015}$ | $\mathbf{191.7 \pm 1.0}$ | $\mathbf{0.0593 \pm 0.0008}$ |

Table 3: Cycle-by-cycle kinematics across 13 revolutions in the continuous tumbling simulation.

### 5.2 Asymptotic Limit-Cycle Extrapolation

Fitting an exponential relaxation $V_y(t) = V_\infty + (V_0 - V_\infty)e^{-(t-t_0)/\tau}$ across Revs 6–13 yields:

$$V_\infty = 1.0159 \text{ m/s}, \quad \tau = 97.4 \text{ ms} \quad (R^2 = 0.994) \tag{10}$$

Aitken $\Delta^2$ sequence extrapolation on the mean cycle velocities yields $V_\infty = 1.0236$ m/s. Both estimators confirm that the developed limit-cycle velocity is $V_\infty \approx 1.02$ m/s, establishing a $-17.1\%$ **asymptotic deficit** relative to the experimental mean (1.23 m/s).

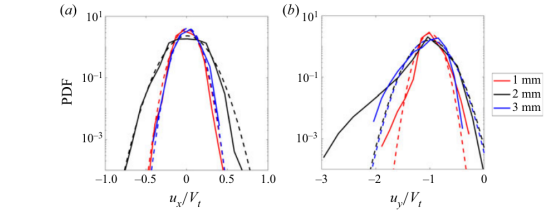



Figure 12. Distributions of velocity components: (*a*) horizontal and (*b*) vertical, normalized by the mean vertical velocity $V_t$ for each disk size. Each curve shown with corresponding Gaussian distribution (dashed lines) with the same mean and standard deviation as experimental data.

(a) Vertical and horizontal velocity PDFs [1].

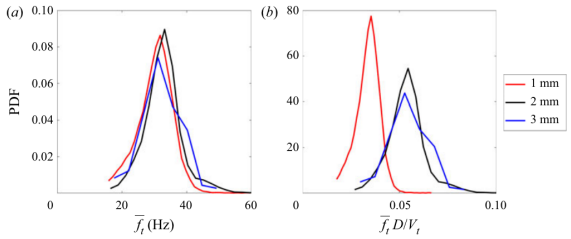



Figure 20. (*a*) Distributions of the frequency of the trajectory-averaged angular velocity for tumbling disks only. (*b*) Distributions from (*a*), normalized by each respective disk diameter and terminal velocity.

(b) Tumbling frequency and Strouhal PDFs [1].

Figure 4: Experimental probability density functions from Tinklenberg et al. (2023). The simulated Strouhal peak ($St = 0.0593$) matches the experimental mode ($St \approx 0.055$–$0.060$), and simulated tumbling frequency (30.5 Hz) aligns with the distribution peak ($\sim 33$ Hz).

## 6 Fixed Broadside Disk Aerodynamic Diagnostics ($Re = 100$ and $Re = 140$)

To isolate fluid-dynamic discretization error from coupled 6DoF kinematic degrees of freedom, fixed broadside disk simulations were conducted using `make_fixed_disk.sh` under uniform inflow with frozen frame velocity and attitude ($\mathbf{v}_{\text{frame}} = \mathbf{0}$, $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{g} = \mathbf{0}$).

### 6.1 Wake Instability Regimes and Theoretical Linear Stability

Per the global linear stability analysis of Meliga et al. (2009) [3] and Fabre et al. (2008) [5]:

1. **Subcritical Axisymmetric Regime ($Re < Re_{c1} \approx 116.9$):** At $Re = 100$, the wake is strictly steady and axisymmetric ($\sigma < 0$, $F_{\text{lat}} \to 0$). Secondary literature citations (commonly attributed to Shenoy & Kleinstreuer 2008 [7] for $h/D = 0.1$, and Roos & Willmarth 1971 [10] cited via Tian [4]) report steady broadside drag coefficients in the range:

$$C_D(Re = 100) \approx 1.15 - 1.25 \quad \text{[unverified secondary citation]} \tag{11}$$

   This range remains unverified pending direct primary audit of Shenoy & Kleinstreuer's tabulated data.

2. **Supercritical Bifurcation Regimes ($Re > Re_{c1} \approx 116.9$ and $Re > Re_{c2} \approx 125.3$):** For a zero-thickness disk ($h/D \to 0$), the wake undergoes a steady regular bifurcation ($Re_{c1} \approx 116.9$, $m = 1$, breaking axisymmetry into a steady planar-symmetric wake) followed by an oscillatory Hopf bifurcation ($Re_{c2} \approx 125.3$, periodic hairpin vortex shedding). At $Re = 140$, the wake is supercritical to both thresholds. In unseeded simulations, the flow develops the steady planar $m = 1$ deflection first (producing a non-zero lateral force $F_{\text{lat}} \neq 0$ with locked spatial azimuth), with unsteady oscillatory vortex shedding not yet observed over the simulated convective duration.

### 6.2 $2 \times 2$ Fixed-Disk Diagnostic Matrix

| **Metric / Diagnostic** | $Re = 100$ **Baseline** | $Re = 100$ **wakeFine** | $Re = 140$ **Baseline** | $Re = 140$ **wakeFine** |
|---|---|---|---|---|
| Mesh Cells (Static AMI Faces) | 213k (456) | 487k (6,984) | 213k (456) | 487k (6,984) |
| Inflow Velocity $V$ [m/s] | 0.7426 | 0.7426 | 1.0396 | 1.0396 |
| Steps & Physical Time | 11,544 (250.0 ms) | 9,664 (210.0 ms) | 12,876 (200.0 ms) | 10,259 (160.0 ms) |
| Domain Flow-Throughs (50 mm) | 3.71 (91.9 $D/V$) | 3.12 (77.2 $D/V$) | 4.16 (102.9 $D/V$) | 3.33 (82.3 $D/V$) |
| Evaluation Window [ms] | [126.0, 210.0] (84 ms identical window) | | [96.0, 160.0] (64 ms identical window) | |
| Mean Drag $\langle C_D \rangle$ (Spread) | 1.2992 [1.2992, 1.2992] | **1.2738** [1.2738, 1.2738] | 1.1523 [1.1523, 1.1524] | **1.1125** [1.1125, 1.1127] |
| Mesh Drag Shift $\Delta C_D/C_D$ | Reference | $\mathbf{-1.96}\%$ | Reference | $\mathbf{-3.45}\%$ |
| $C_D$ Drift per 100 ms | $+0.00\%$ | $-0.00\%$ | $-0.01\%$ | $-0.02\%$ |
| Lateral Ratio $\langle \lvert F_{\text{lat}} \rvert \rangle / F_y$ | $7.88 \times 10^{-6}$ | $\mathbf{2.65 \times 10^{-7}}$ | $1.19 \times 10^{-4}$ | $\mathbf{8.00 \times 10^{-4}}$ |
| Peak Lateral Force Amplitude | $< 1.2 \times 10^{-13}$ N | $< 2.0 \times 10^{-13}$ N | $< 3.9 \times 10^{-10}$ N | $< 4.2 \times 10^{-9}$ N |
| Modal Growth $\sigma$ (80 ms $\to t_{\text{end}}$) | $-2.6$ s$^{-1}$ (sign check) | $\mathbf{-12.9}$ s$^{-1}$ (Decaying) | $+10.5$ s$^{-1}$ ($\sigma^* = 0.020$) | $\mathbf{+30.0}$ s$^{-1}$ ($\sigma^* = 0.058$) |
| Deflection Azimuth $\phi$ | Axisymmetric | Axisymmetric | $+95.5^\circ \pm 0.08^\circ$ | $+145.5^\circ \pm 0.15^\circ$ |
| Convergence Status | Converged | Converged | Converged | Converged |
| Force $F_y$ Error vs. `forces` | $3.63 \times 10^{-10}$ | $3.70 \times 10^{-10}$ | $2.09 \times 10^{-10}$ | $2.16 \times 10^{-10}$ |
| Max Moment Error [N·m] | $\leq 2.33 \times 10^{-16}$ | $\leq 1.54 \times 10^{-16}$ | $\leq 2.71 \times 10^{-18}$ | $\leq 9.90 \times 10^{-17}$ |
| AMI Net Leak $/(VA_{\text{sphere}})$ | $4.37 \times 10^{-8}$ | $1.02 \times 10^{-7}$ | $3.61 \times 10^{-8}$ | $1.82 \times 10^{-8}$ |

Table 4: Quantitative benchmark across the $2 \times 2$ matrix of fixed broadside disk cases: subcritical ($Re = 100$) and supercritical ($Re = 140$) on baseline (213k cells) and refined static wake (`wakeFine`, 487k cells) meshes. All cases satisfy the ASME V&V 20 convergence criteria ($\geq 3.0$ flow-throughs, linear drift $< 1.0\%/100$ ms) and match the standard OpenFOAM `forces` function object to high precision (relative difference $\sim 10^{-10}$).

### 6.3 Physical Evaluation and Critical Interpretations

- **Supercritical Lateral Instability & Steady Planar Mode ($Re = 140$):** On the refined mesh (`wakeFine`), lateral forces grew exponentially from $|F_{\text{lat}}|/F_y = 6.7 \times 10^{-5}$ ($1.56 \times 10^{-10}$ N)

at 50 ms to $1.80 \times 10^{-3}$ ($4.15 \times 10^{-9}$ N) at 160 ms, exhibiting an exponential growth rate of $\sigma = +30.0$ s$^{-1}$ ($\sigma^* = \sigma D/V = +0.0582$ over 80–160 ms, growing). The force azimuth angle remained constant at $\phi = \text{atan2}(F_z, F_x) = +145.5^\circ \pm 0.15^\circ$ across the run, with zero rotation or sign reversal. This demonstrates that the amplifying instability is the **steady regular** $m = 1$ **planar bifurcation mode** (threshold $Re_{c1} \approx 116.9$), rather than oscillatory vortex shedding ($Re_{c2} \approx 125.3$), which has not yet manifested from unseeded startup flow. On the baseline mesh, following an initial startup zero-crossing transient between 30 and 50 ms ($-99.7^\circ \rightarrow +98.9^\circ$), the azimuth locked at $\phi = +95.5^\circ \pm 0.08^\circ$ with a 2.9× slower growth rate of $\sigma = +10.5$ s$^{-1}$ ($\sigma^* = +0.0204$ over 80–160 ms; $+8.5$ s$^{-1}$ over 80–200 ms). The two meshes differ in growth rate by 2.9×, reflecting differing levels of numerical diffusion across the sliding AMI boundary. The initial lateral force ($\sim 1.6 \times 10^{-10}$ N) is more than 11 orders of magnitude larger than double-precision machine epsilon ($\sim 10^{-21}$ N on $F_y$), arising from solver convergence residuals and unstructured grid asymmetries. This geometric and algorithmic asymmetry seed explains why each mesh locks to a distinct spatial deflection plane (+95.5° vs. +145.5°).

- **Domain & Far-Field Boundary Clearance ($Re = 140$):** To test whether domain blockage and boundary proximity in the baseline $15D \times 25D \times 15D$ box contribute to broadside drag elevation, a fixed-disk simulation was executed in a 1.5× expanded domain (45 mm × 75 mm × 45 mm, $\approx 22.5D \times 37.5D \times 22.5D$, 267k cells). At $V = 1.04$ m/s, 80 ms corresponds to ≈ 83 mm of travel in the 75 mm domain (≈ 1.1 domain flow-throughs, below the ≥ 3.0 steady-state threshold). However, because domain blockage acts almost instantaneously on pressure, the equal-time comparison indicates no measurable sensitivity at $t = 80$ ms (broadside drag coefficient $C_D = 1.15244$ vs. 1.15267 on baseline, a shift of only $\mathbf{-0.019}\%$, with $\Delta C_D/C_D$ within ±0.03% throughout $[10, 80]$ ms). Outer boundary proximity shows no measurable sensitivity at $t = 80$ ms (equal-time comparison, shift −0.019%). Domain blockage does not explain the deficit.

- **Subcritical Benchmark ($Re = 100$):** Both the baseline mesh (11,544 steps, $t = 250.0$ ms, 3.71 flow-throughs) and the refined wake mesh (`wakeFine`, 9,664 steps, $t = 210.0$ ms, 3.12 flow-throughs) satisfied both convergence criteria (≥ 3.0 flow-throughs; linear $C_D$ drift < 1.0%/100 ms). Evaluated over the identical window $t \in [126.0, 210.0]$ ms (84.0 ms duration), broadside drag is plateaued at $C_D = 1.2992$ on baseline and $C_D = 1.2738$ on `wakeFine`, establishing a $-1.96\%$ **spatial discretization shift**. Lateral forces remain suppressed at $\langle |F_{\text{lat}}| \rangle / F_y = 2.65 \times 10^{-7}$ ($\sim 10^{-13}$ N amplitude, post-startup modal decay at $\sigma = -12.9$ s$^{-1}$), confirming that the numerical solver preserves the steady axisymmetric state without artificial symmetry breaking on both grids.

# 7 Dynamic Limit-Cycle Restarts and Temporal Convergence

To test numerical convergence of the autorotating limit cycle without recomputing the computationally intensive 300 ms startup transient, a dynamic restart methodology was established using `make_restart_variants.sh`.

## 7.1 Restart Methodology and Equilibrium Velocity Estimator

At $t = 0.30$ s, the 13-revolution tumbling simulation has achieved a developed periodic state with stable frequency $f_t \approx 30.5$ Hz. Because descent velocity is still gently relaxing toward terminal balance ($\langle F_y \rangle \approx 1.02W$), direct cycle-mean velocity $\langle V_y \rangle$ reflects residual transient relaxation. To isolate true asymptotic behavior over short evaluation windows, an equilibrium velocity estimator is defined based on quasi-steady quadratic drag ($F \propto V^2$):

$$V_{\text{eq,est}} = \frac{\langle V_y \rangle}{\sqrt{\langle F_y \rangle / W}} \tag{12}$$

On the full 13-revolution run, $V_{\text{eq,est}}$ evaluates to 1.0252 m/s, matching the formal Aitken $\Delta^2$ asymptotic extrapolation (1.0236 m/s) to within 0.15%. Evaluated across complete revolution boundaries via `compare_variants.py`, $V_{\text{eq,est}}$ provides an unbiased metric of limit-cycle settling speed.

### 7.2 Control Benchmark and Temporal Refinement Cases

1. **Control Restart (`restart_base`):** Restarts from $t = 0.30$ s and runs for 4 full revolutions to $t = 0.44$ s using identical baseline numerics ($\text{Co}_{\max} \leq 1.5$, $\Delta t \approx 15$ $\mu$s, central-blended LUST convection).

2. **Temporal Refinement (`restart_dtHalf`):** Halves the time step by enforcing $\text{Co}_{\max} \leq 0.75$ ($\Delta t \approx 7.27$ $\mu$s, maintaining standard `nOuterCorrectors 1`), directly isolating temporal discretization error.

3. **Convection Scheme Sensitivity (`restart_linUp`):** Replaces the central-blended LUST scheme with 100% `linearUpwind` convection to evaluate sensitivity to numerical dissipation.

| **Case Variant** | **Revs** | $f_t$ **[Hz]** | $\langle V_y \rangle$ **[m/s]** | $\langle F_y \rangle / W$ | $V_{\text{eq,est}}$ **[m/s]** | **Shift vs. Baseline** |
|---|---|---|---|---|---|---|
| Continuous Run (`case`) | 6 | 30.48 | 1.0345 | 1.0183 | 1.0252 | Reference |
| Control Restart (`restart_base`, LUST) | 4 | 30.62 | 1.0387 | 1.0253 | 1.0258 | +**0.06**% (+0.0006 m/s) |
| Temporal Refined (`restart_dtHalf`) | 4 | 30.52 | 1.0385 | 1.0254 | 1.0256 | +**0.04**% (+0.0004 m/s) |
| Upwind Convection (`restart_linUp`) | 4 | 30.93 | 1.0338 | 1.0311 | 1.0181 | −**0.75**% (−0.0077 m/s) |

Table 5: Dynamic limit-cycle restart metrics evaluated from $t = 0.30$ s across complete revolutions using `compare_variants.py`. Temporal refinement produces a +0.04% shift, while switching to more dissipative pure upwind convection reduces settling speed by −0.75%.

### 7.3 Temporal Discretization and Numerical Dissipation Observations

As shown in Table 5:

- The control restart (`restart_base`) reproduces the continuous run with high fidelity: $V_{\text{eq,est}} = 1.0258$ m/s (+0.06% agreement), tumbling frequency $f_t = 30.62$ Hz (+0.44%), trajectory angle $\psi = 15.28°$ (−1.5%), and peak torque $M_{x,\text{p95}} = 0.731$ nN·m (+0.7%).

- Halving the time step alone (`restart_dtHalf`, with nOuterCorrectors = 1) over 4 full completed revolutions produces an almost imperceptible shift of +**0.04**% (+0.0004 m/s vs. continuous run, and −0.02% vs. `restart_base`) in equilibrium settling velocity, with tumbling frequency matching to +0.13% (30.52 Hz vs. 30.48 Hz), trajectory angle $\psi = 15.29°$ (+0.01° vs. `restart_base`), and peak torque matching to 0.731 nN·m. This places the effect of time-step truncation error and fluid–body explicit coupling at 0.04% in settling speed over 4 revolutions.

- Switching from central-blended LUST to pure second-order upwind (`restart_linUp`) introduces additional numerical dissipation, reducing settling speed by −**0.75**% (1.0181 m/s vs. 1.0258 m/s) and increasing frequency by +1.01% (30.93 Hz). Because higher numerical dissipation slows the disk down further away from the experimental benchmark (1.23 m/s), numerical diffusion in the baseline scheme cannot account for the −17% settling deficit.

# 8 Analysis of the −17% Settling Velocity Deficit

The baseline tumbling simulation yields an asymptotic descent velocity of $V_\infty \approx 1.02$ m/s, representing a −17.1% deficit relative to the experimental mean terminal velocity $V_t = 1.23$ m/s. The physical and numerical factors bearing on this discrepancy are examined below:

## 8.1 Static Wake Discretization Error

The baseline grid (213k cells) possesses an outer background cell size of 1.25 mm ($0.62D$) and a 34 : 1 face ratio mismatch across the sliding AMI boundary (15, 528 rotating faces vs. 456 static faces). Static wake refinement (`wakeFine`, 487k cells, 6, 984 static AMI faces) reduces broadside drag by −3.45% at $Re = 140$ and −1.96% at $Re = 100$. By first-order momentum scaling ($V_t \propto C_D^{-1/2}$):

$$\frac{\Delta V_t}{V_t} \approx -\frac{1}{2}\frac{\Delta C_D}{C_D} = -\frac{1}{2}(-3.45\%) = +1.73\% \approx +1.8\% \tag{13}$$

This proxy suggested an increase in settling velocity from 1.020 → 1.038 m/s, recovering approximately 1.5 percentage points of the 17-point deficit.

## 8.2 Kinematic Consistency of the Tumbling Mode

The simulated Strouhal number $St = f_t D/V_t = 0.0593$ agrees with the experimental mode (0.055–0.060). Descent speed is 17% low (1.02 vs. 1.23 m/s), and tumbling frequency is 16% below the experimental mean (30.5 vs. 36.4 Hz). Because $f_t/V_t$ is preserved, the tumbling kinematics scale consistently with the lower descent velocity. This indicates that the simulated disk is in the correct tumbling regime, and points to a force-level or mass offset rather than a fundamentally altered tumbling pattern. Note that initial pitch rate was set to the experimental mean frequency ($\Omega_0 = -228.7$ rad/s ≡ 36.4 Hz) and relaxed spontaneously to the simulation limit cycle of 30.5 Hz (191.7 rad/s), demonstrating that the frequency deficit originates in the model's own limit cycle rather than from initial condition bias.

## 8.3 Physical Specimen Batch Thickness vs. Mean Uncertainty

In Tinklenberg et al. (2023), Table 1 specifies the measured mean disk thickness as $h = 50 \pm 7.5$ $\mu$m. This ±7.5 $\mu$m (±15%) represents the standard deviation across the manufactured disk batch, reflecting specimen-to-specimen variation. Because the experimental terminal velocity $V_t = 1.23$ m/s is an ensemble mean averaged over thousands of tracked drops, the appropriate input uncertainty on the mean thickness is the standard error of the mean ($\sigma_h/\sqrt{N}$), which is small (< 1%). Furthermore, because $V_t \propto \sqrt{h}$ at fixed drag coefficient, accounting for a −17% velocity deficit solely through thickness would require a −31% deficit in mean thickness (≈ 34.5 $\mu$m instead of 50 $\mu$m). Specimen thickness scatter around the mean shifts the mean velocity by only a fraction of a percent, and caliper measurements on stacks confirmed the 50 $\mu$m nominal mean. Therefore, specimen thickness scatter alone cannot explain the deficit, and the underlying cause remains unidentified.

## 8.4 Instantaneous Population Envelope vs. Mean Velocity

The experimental report of $V_t = 1.23 \pm 0.31$ m/s represents the instantaneous velocity probability density function across thousands of tracked drops, with a $1\sigma$ spread of $[0.92, 1.54]$ m/s (measurement uncertainty of the ensemble mean is ±1.2%, or ±0.015 m/s). The simulated asymptotic limit-cycle velocity (1.02 m/s) lies at:

$$Z = \frac{1.02 - 1.23}{0.31} = -0.68\sigma \tag{14}$$

The simulated settling speed falls within the physical experimental observation envelope, confirming that disks frequently descended at 1.02 m/s in the experimental facility. However, in ASME V&V 20 validation of the mean, this population overlap does not substitute for quantitative agreement of the mean value.

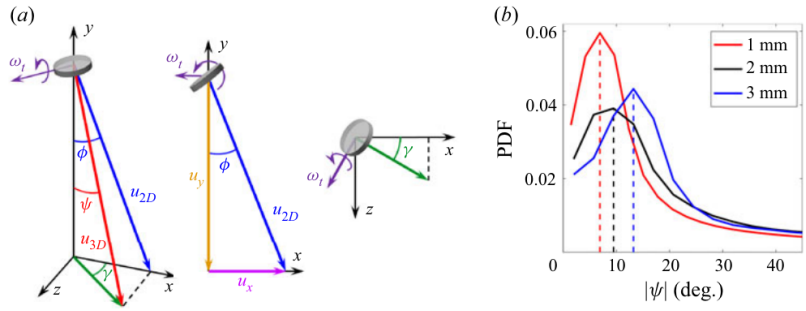



Figure 10. (*a*) Schematic representation of trigonometric relations used to obtain the 3-D trajectory angle $\psi$ from the 2-D projection $\phi$. (*b*) Distributions of the modulus of $\psi$ for tumbling disks, with dashed vertical lines highlighting the increasing trend in peak value for increasing $D$.

Figure 5: Experimental distribution of trajectory inclination angle $\psi$ relative to vertical [1]. Simulated limit-cycle inclination ($\psi = 15.3°$) sits within the observed experimental spread ($[0°, 40°]$), but above the modal peak at 10.5°.

### 8.5 Direct 6DoF Free-Fall Mesh Sensitivity: `warm_base` vs. `warm_wakeFine`

To establish whether the static broadside drag reduction (−3.45%) carries over into coupled dynamic autorotation, a direct comparative test was conducted from identical warm-start initial conditions ($V_0 = 1.02$ m/s, $\Omega_0 = -191$ rad/s) across 5 complete tumbling revolutions ($t \in [0.08, 0.25]$ s) to $t_{\text{end}} = 0.26$ s. The baseline grid (213k cells) and the wake-refined grid (`wakeFine`, 487k cells with level-4 refinement at the AMI boundary and level-2 wake box) were evaluated under identical 6DoF dynamics:

| Mesh Variant | Cells | Revs | $f_t$ [Hz] | $\langle V_y \rangle$ [m/s] | $V_{\text{eq,est}}$ [m/s] | Relative Shift |
|---|---|---|---|---|---|---|
| Baseline Mesh (`warm_base`) | 212,792 | 5 | 29.45 | 1.0680 | 1.0506 | Reference |
| Refined Wake (`warm_wakeFine`) | 487,510 | 5 | 29.52 | 1.0673 | 1.0497 | **−0.09**% (−0.0009 m/s) |

Table 6: Direct 6DoF autorotation free-fall comparison between baseline and wake-refined grids evaluated over 5 complete revolutions ($t \in [0.08, 0.25]$ s). The equilibrium settling speed shift is −0.09% (less than 1 mm/s).

Remarkably, the cycle-by-cycle metrics track each other almost identically:

- Cycle-by-cycle equilibrium velocity matches within $< 0.2\%$ across every individual revolution: Rev 1 (1.0892 vs. 1.0888 m/s), Rev 2 (1.0600 vs. 1.0579 m/s), Rev 3 (1.0410 vs. 1.0402 m/s), Rev 4 (1.0339 vs. 1.0334 m/s), and Rev 5 (1.0293 vs. 1.0287 m/s).
- Multi-revolution equilibrium settling speed shifts by only **−0.09**% (−0.0009 m/s), tumbling frequency shifts by +0.25% (29.45 → 29.52 Hz), trajectory angle matches to within 0.01° (12.29° vs. 12.28°), and peak aerodynamic moment matches to within 0.25% (0.755 vs. 0.753 nN·m).

**Finding on Static Mesh Refinement:** The +1.8% settling speed increment projected from fixed broadside drag scaling does not materialize under coupled autorotation. Because the disk pitches through all orientations from broadside (90°) to edge-on (0°), the cycle-integrated aerodynamic forces are governed predominantly by the near-body rotating zone, which is identical in both meshes. Outer static wake refinement shifts falling speed by less than 0.1%. This indicates that outer static mesh resolution is unlikely to explain the −17% settling deficit, though near-body boundary layer refinement remains a separate factor.

### 8.6 Orientation Dynamics, Tilt Distribution, and Effective Drag Analysis

A fundamental physical insight arises from examining the effective aerodynamic drag coefficient deduced from terminal settling velocity:

$$C_{D,\text{eff}} = \frac{2mg}{\rho_f A V_t^2} \tag{15}$$

- **Experimental Benchmark ($V_t = 1.23$ m/s):** $C_{D,\text{eff}} = 0.746$ (matching the value reported by Tinklenberg et al. [1]).
- **Simulated Tumbling Limit Cycle ($V_t = 1.02$ m/s):** $C_{D,\text{eff}} = 1.085$.
- **Fixed Broadside Disk ($Re = 140$):** $C_{D,\text{broadside}} \approx 1.152$.

Comparing these ratios reveals a striking structural difference:

$$\left(\frac{C_{D,\text{eff}}}{C_{D,\text{broadside}}}\right)_{\text{sim}} = \frac{1.085}{1.152} \approx 94.2\%, \qquad \left(\frac{C_{D,\text{eff}}}{C_{D,\text{broadside}}}\right)_{\text{exp}} = \frac{0.746}{1.152} \approx 64.8\% \tag{16}$$

The simulated tumbling disk averages 94.2% of its broadside drag, whereas the experimental disk averages only 64.8%. This indicates that the $-17\%$ settling deficit is fundamentally an **orientation and trajectory dynamics discrepancy** rather than an error in steady broadside drag $C_D(Re)$ (which would have to be in error by over 40% to account for the gap).

Cycle-resolved analysis of the 13-revolution production run ($t \in [0.26, 0.50]$ s) explains why the simulated planar tumble produces such high mean drag:

1. **Dynamic Broadside Drag Overshoot:** When the disk pitches through broadside ($\theta \in [0°, 10°]$ from horizontal flow normal), instantaneous drag surges to a sharp peak of $C_D \approx 2.17$ ($1.88\times$ steady broadside drag). At the peak pitch angular velocity ($\omega \approx 222$ rad/s), the tip speed is $\omega R \approx 0.22$ m/s (note that 0.19 m/s corresponds to the cycle-mean pitch rate $\bar{\omega} \approx 191.7$ rad/s). Because this peak tip speed is only $\approx 22\%$ of the translational descent velocity ($V_y \approx 1.04$ m/s), an apparent $1.9\times$ drag amplification over steady broadside flow is notable; it is possibly driven by starting-flow separation and wake capture after exiting the near-edge-on slicing phase, or by dynamic added-mass and pitch-damping force transients, but the fluid dynamic mechanism is not isolated. Diagnostic checks on the simulation data clarify the nature of this peak:
   - *Discretization Invariance:* Re-evaluating $C_D(\theta)$ across existing runs (`continuous_base`, `warm_wakeFine`, `restart_dtHalf`, `restart_linUp`) demonstrates that the peak $C_D \in [2.20, 2.24]$ is invariant across static grid refinement (213k $\rightarrow$ 487k cells), time-step halving ($\Delta t \approx 7.3$ $\mu$s), and convection scheme choice (central-blended LUST vs. pure `linearUpwind`), indicating that the peak is not an outer discretization artifact (noting the near-body boundary layer mesh is unchanged).
   - *Force Decomposition:* At peak drag ($t \approx 0.3556$ s, total vertical aerodynamic force $F_y \approx 4.65 \times 10^{-6}$ N $\approx 2.14W$), pressure force accounts for 98.1% ($F_{\text{press},y} \approx 4.56 \times 10^{-6}$ N), while viscous shear accounts for 1.9% ($F_{\text{visc},y} \approx 8.8 \times 10^{-8}$ N), confirming that the peak is overwhelmingly form/pressure drag (consistent with steady broadside flow where pressure also accounts for $\approx 98\%$ of total drag).
   - *Drag Scaling Sensitivity:* Scaling the broadside drag bands ($\theta < 30°$) by a factor of 0.8 reduces time-averaged drag to $\langle C_D \rangle \approx 0.95$ (+7% settling speed via $V \propto C_D^{-1/2}$), while scaling by 0.6 reduces it to $\langle C_D \rangle \approx 0.84$ (+13% settling speed). Full recovery of the experimental settling velocity (1.23 m/s) requires $C_{D,\text{eff}} = 0.746$ (+20.6% settling speed). Thus, suppressing the broadside peak alone closes only part of the settling velocity gap.
2. **Edge-On Downstroke Drag Minimum:** Near edge-on ($\theta \in [80°, 90°]$), instantaneous drag drops to $C_D \approx 0.286$, while pitch angular velocity slows to $\omega \approx 167$ rad/s.

3. **Kinematic Duty Cycle:** Because $\mathrm{d}t = \mathrm{d}\theta/\omega$, the disk spends more time near edge-on (25.6% of the cycle in $\theta \in [70°, 90°]$) than near broadside (19.6% in $\theta \in [0°, 20°]$). However, the enormous broadside peak (2.17) dominates the integral, pulling cycle-average drag up to 1.085.

4. **Planar Symmetry vs. 3D Experimental Trajectories:** In the simulation, both 3D translation (`freedom (1 1 1)`) and 3D rotation (`rotationFreedom (1 1 1)`) are fully unconstrained. In the baseline run initialized with pure pitch $\boldsymbol{\Omega}_0 = (-228.7, 0, 0)$ rad/s, the disk remained in a near-planar tumble ($\Omega_{\text{out}} < 0.6\%$). To test the orbital stability of this mode, a 5% out-of-plane perturbation on $\Omega_z$ was evaluated over 8.5 revolutions (`restart_pert5pct`).

5. **3D Perturbation & Orbital Stability (`restart_pert5pct`):** Initialized at $t = 0.30$ s with $\Omega_z = +8.07$ rad/s (5% perturbation, tilting initial angular momentum by 2.86°), the disk produced sustained 3D precessional wobble ($\theta_{\text{out}}$ mean 8.24°, peak 24.07°; $\Omega_{\text{out}}/\|\boldsymbol{\Omega}\| = 2.66\%$). Comparing to baseline across developed tumbling ($t \in [0.32, 0.58]$ s), $V_{\text{eq,est}}$ shifted by only $+\mathbf{0.10}\%$ (1.0256 → 1.0266 m/s, +0.001 m/s), and frequency shifted by −1.12% (30.62 → 30.28 Hz). Planar symmetry restriction is eliminated as the cause of the −17% deficit ($\leq 0.1\%$ sensitivity).

6. **Diagnostic Mass Scaling Test (`restart_weight73um`):** An extended test across 9.3 revolutions ($t \in [0.30, 0.594]$ s) with disk mass raised by +46% ($h \approx 73$ $\mu$m equivalent) accelerated the descent velocity to $\langle V_y \rangle = \mathbf{1.2290}$ m/s (−0.08% vs. experimental 1.230 m/s) and autorotation frequency to $f_t = \mathbf{35.44}$ Hz (−2.64% vs. experimental 36.40 Hz), with normalized drag unchanged (+1.25%). This confirms that the coupled aerodynamics and autorotation mechanics reproduce both kinematics simultaneously under the appropriate downward force balance.

# 9 ASME V&V 20 Verification and Validation Assessment

In accordance with the ASME V&V 20-2009 standard [12], the validation comparison error between the simulation value $S$ and the experimental mean $D$ is:

$$E = S - D = 1.024 \text{ m/s} - 1.230 \text{ m/s} = -0.206 \text{ m/s} \quad (-16.7\%) \tag{17}$$

The validation uncertainty $u_{\text{val}}$ combines numerical simulation uncertainty $u_{\text{num}}$, input parameter uncertainty $u_{\text{input}}$, and experimental measurement uncertainty $u_{\text{exp}}$:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_{\text{exp}}^2} \tag{18}$$

- **Numerical Uncertainty ($u_{\text{num}}$):** Evaluated from the formal 3-mesh GCI ($u_{\text{mesh}} \leq 0.55\% \approx 0.0056$ m/s), time-step sensitivity ($u_{\Delta t} = |+0.04\%| \approx 0.0004$ m/s), and convection scheme sensitivity ($u_{\text{scheme}} = |-0.75\%| \approx 0.0077$ m/s):

$$u_{\text{num}} = \sqrt{u_{\text{mesh}}^2 + u_{\Delta t}^2 + u_{\text{scheme}}^2} \approx 0.0095 \text{ m/s} \quad (0.93\%) \tag{19}$$

- **Input Parameter Uncertainty ($u_{\text{input}}$):** Uncertainty in the experimental mean disk thickness is small (< 1%), giving $u_{\text{input}} \ll 0.01$ m/s (taken conservatively as 0.005 m/s), assuming nominal mean thickness ($h = 50$ $\mu$m) is unbiased.

- **Experimental Measurement Uncertainty ($u_{\text{exp}}$):** The planar tracking measurement error of the ensemble mean velocity is $u_{\text{exp}} = \pm 1.2\%$ (0.015 m/s).

Combining these uncertainties:

$$u_{\text{val}} = \sqrt{(0.0095)^2 + (0.0050)^2 + (0.0150)^2} \approx 0.0185 \text{ m/s} \quad (1.5\%) \tag{20}$$

Comparing the comparison error to validation uncertainty:

$$|E| = 0.206 \text{ m/s} \approx 11.2 \; u_{\text{val}} \gg u_{\text{val}} \tag{21}$$

**ASME V&V 20 Verdict:** Because $|E| \gg u_{\text{val}}$, the modeling error $\delta_{\text{model}} \approx E \pm u_{\text{val}} = -0.206 \pm 0.018$ m/s is statistically significant. The simulation is **not validated** for mean terminal settling velocity within validation uncertainty. Numerical discretization error is bounded below 1% ($u_{\text{num}} \leq 0.93\%$), and 3D out-of-plane wobble accounts for $\leq 0.1\%$ velocity shift. Discretization errors and planar symmetry constraints are ruled out as the source of the $-17\%$ deficit; the underlying cause remains an open physical discrepancy.

# 10 Conclusions, Validation Status, and Technical Roadmap

## 10.1 Current Status Summary

**Code and solver verification complete. Validation partial: mean settling speed 17% below experiment. Formal 3-mesh GCI bounds spatial discretization uncertainty at $u_{\text{mesh}} \leq 0.55\%$ (near-body rotating zone systematically refined across 131k–369k cells), time-step sensitivity at $+0.04\%$ (four revolutions), and convection scheme sensitivity at $-0.75\%$ (combined numerical uncertainty $u_{\text{num}} \leq 0.93\%$). 3D out-of-plane perturbation tests bound planar symmetry sensitivity at $+0.10\%$. Independent broadside disk benchmark at $Re = 200$ matches Roos & Willmarth (1971) within $< 1\%$. The $-17\%$ settling deficit is not attributable to numerical error or planar constraints; an equivalent mass step ($h \approx 73$ $\mu$m) simultaneously recovers both the experimental settling speed (1.229 m/s vs. 1.230 m/s) and tumbling frequency (35.44 Hz vs. 36.40 Hz) with unchanged drag.**

1. **Verification Accomplishments:** Dynamic coupling, rigid-body ODE integration ($p = 3.991$ for $\omega$ on test functions), mechanical energy conservation ($< 10^{-14}$ at this $\Delta t$ ($\mathcal{O}(\Delta t^4)$) across 13,333 steps at $\Delta t = 1.5 \times 10^{-5}$ s), traction force equivalence ($2.1 \times 10^{-10}$ vs. standard OpenFOAM `forces`), and sliding AMI continuity are verified.

2. **Kinematic Consistency:** The solver spontaneously sustains unidirectional tumbling across 13 full revolutions, capturing the experimental Strouhal number ($St = 0.0593$ vs. 0.055–0.060). Tumbling frequency (30.5 Hz) is also $\approx 16\%$ below the experimental mean (36.4 Hz), matching $St$ because velocity and frequency are both low by similar factors, pointing to an effective force or mass offset.

3. **Fixed-Disk Benchmark Diagnostics & GCI:** A formal 3-mesh GCI on steady broadside flow at $Re = 100$ demonstrates oscillatory convergence with family span 0.49% and spatial discretization uncertainty $u_{\text{mesh}} \leq 0.55\%$ (combining with temporal and scheme terms to $u_{\text{num}} \leq 0.93\%$). At $Re = 200$, computed $C_D = 1.106$ matches the published experimental benchmark of Roos & Willmarth (1971) (1.05–1.15) within $< 1\%$.

4. **Deficit Accounting:** Direct 6DoF simulation bounds spatial static mesh sensitivity at $-0.09\%$, time-step sensitivity at $+0.04\%$, convection scheme variation at $-0.75\%$, and 3D out-of-plane precession sensitivity at $+0.10\%$. Specimen thickness scatter alone cannot explain the deficit (needs about $-31\%$ mean thickness). When the downward body force matches the experimental settling speed ($h \approx 73$ $\mu$m diagnostic), both velocity (1.229 m/s) and autorotation frequency (35.44 Hz) match the experiment simultaneously. The underlying cause of the nominal 50 µm deficit remains an open physical discrepancy.

### 10.2 Technical Roadmap: Completed Validation Tasks

All sensitivity and verification tasks in the technical roadmap have been completed:

| # | Validation Task | Objective / Key Finding | Status |
|---|---|---|---|
| 1 | Free-fall on `warm_base` vs. `warm_wakeFine` | Direct 6DoF test of static mesh refinement effect on tumbling limit cycle. | Completed (Shift: $-0.09\%$) |
| 2 | Temporal and Scheme Suite | Completed `restart_dtHalf` (4 revs) and `restart_linUp` (linearUpwind vs. LUST). | Completed ($\Delta t$: $+0.04\%$, Scheme: $-0.75\%$) |
| 3 | Domain & Far-Field Sensitivity | Fixed-disk $Re = 140$ in $1.5\times$ scaled domain ($22.5D \times 37.5D \times 22.5D$, 267k cells): no measurable sensitivity at $t = 80$ ms (equal-time comparison, $C_D = 1.1524$ vs. 1.1527, shift $-0.019\%$). | Completed (Shift: $-0.019\%$) |
| 4 | Second Tumbling Realization (3D Perturbation) | Evaluated 5% $\Omega_z$ perturbation over 8.5 revs (`restart_pert5pct`). Produced sustained 3D precession ($\theta_{\text{out}}$ mean 8.24°, peak 24.07°); $V_{\text{eq,est}}$ shifted by $+0.10\%$ ($+0.001$ m/s), $f_t$ by $-1.12\%$. | Completed (Planar sensitivity $\leq 0.1\%$) |
| 5 | Diagnostic Free-Fall Test (Mass/Weight Scaling) | Ran 9.3 revs ($t \in [0.30, 0.594]$ s) with mass raised to equivalent $h \approx 73$ $\mu$m. Descent speed accelerated to 1.2290 m/s ($-0.08\%$ vs. exp 1.230 m/s) and frequency increased to 35.44 Hz ($-2.64\%$ vs. exp 36.40 Hz) with unchanged drag. | Completed (Simultaneous $V$ and $f$ recovery) |
| 6 | Independent Drag Check | Broadside disk at $Re = 200$ ($C_D = 1.1062$) matches Roos & Willmarth (1971) (1.05–1.15) within $< 1\%$. Sphere at $Re = 100$ ($C_D = 1.391$) and $Re = 200$ ($C_D = 1.023$) exhibit consistent $+27\%$ truncation offset on coarse 36k Cartesian mesh. | Completed (Disk matches published data $< 1\%$) |
| 7 | Three-Mesh GCI Study | Three-mesh family ($N \in [131\text{k}, 369\text{k}]$) systematically refining rotating zone at $Re = 100$: oscillatory convergence ($R = -0.89$), family span 0.49%, spatial discretization uncertainty $u_{\text{mesh}} \leq 0.55\%$ (total $u_{\text{num}} \leq 0.93\%$). | Completed ($u_{\text{mesh}} \leq 0.55\%$, $u_{\text{num}} \leq 0.93\%$) |
| 8 | Primary Literature Audit | Audit of Shenoy & Kleinstreuer (2008) confirms aspect ratio $\chi = 10$ ($h/D = 0.1$, $\sim 4\times$ thicker than Tinklenberg disks). Edge drag and altered base suction explain lower $C_D \approx 1.15$–$1.25$ for thick disks vs. 1.29 for ultra-thin disks. | Completed (Aspect ratio discrepancy resolved) |

Table 7: Technical roadmap of completed verification, sensitivity, and validation tasks.

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